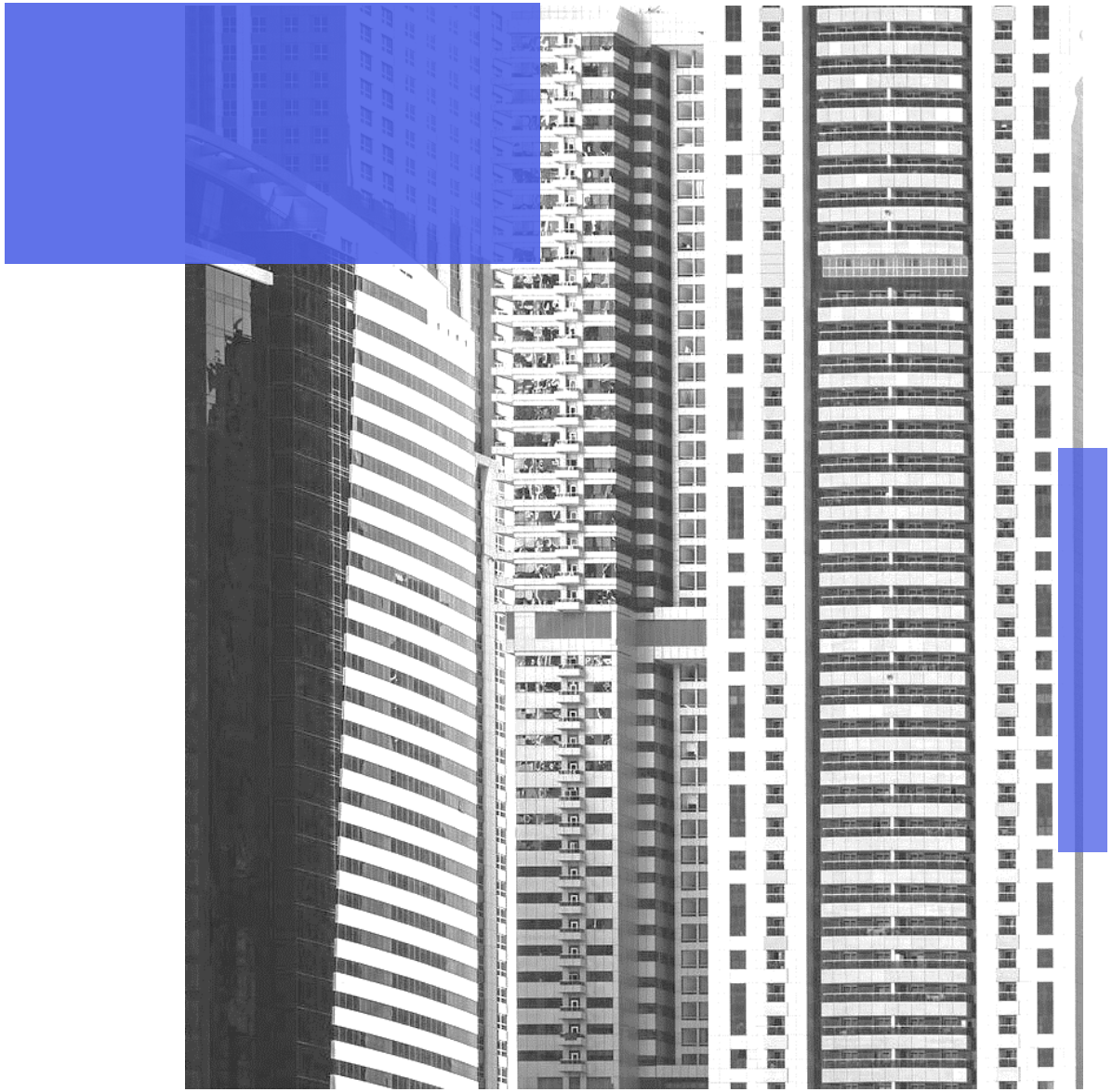
MARKET ANALYSIS OF

REAL ESTATE





AGENDA

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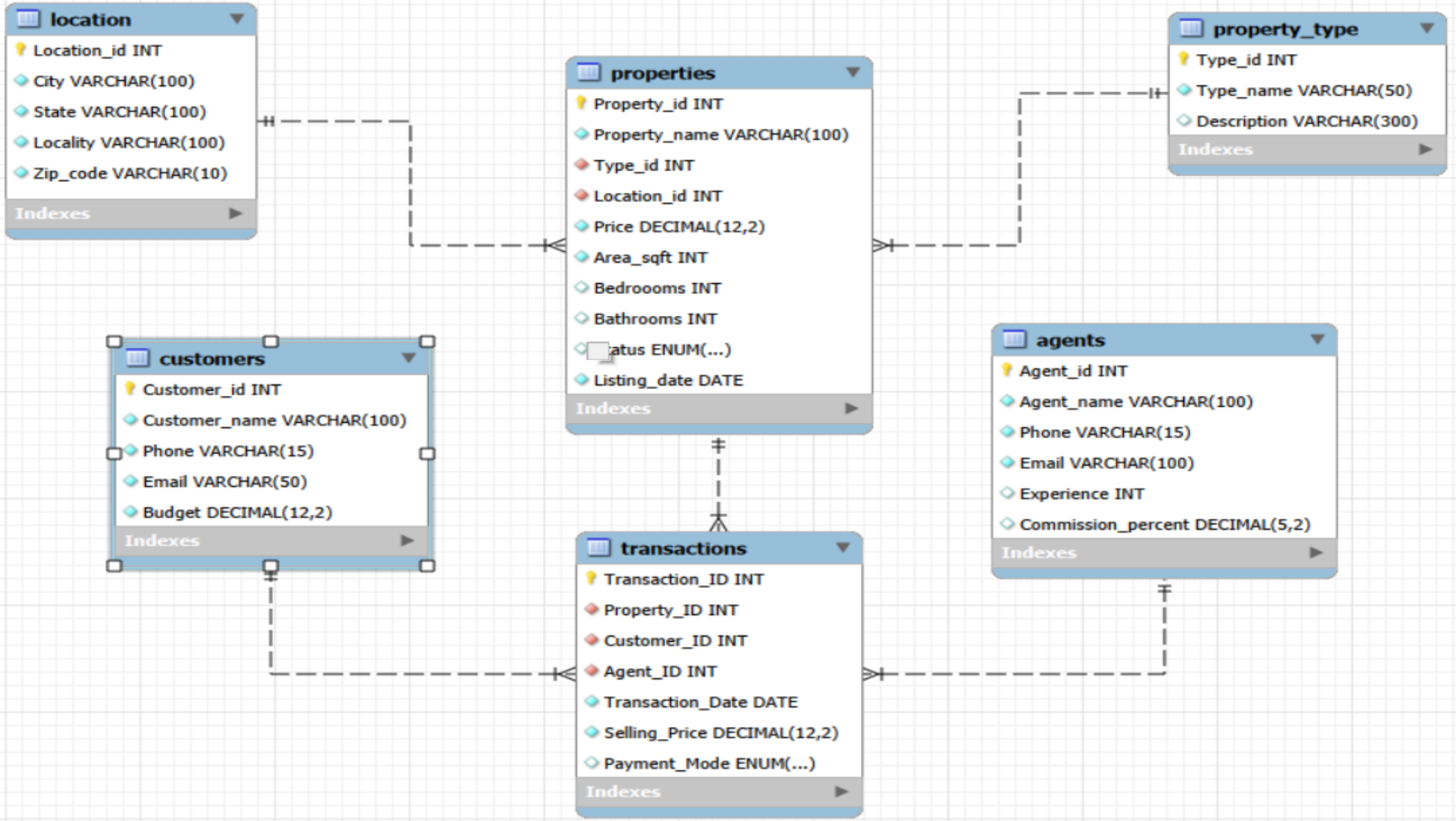
VIEWS

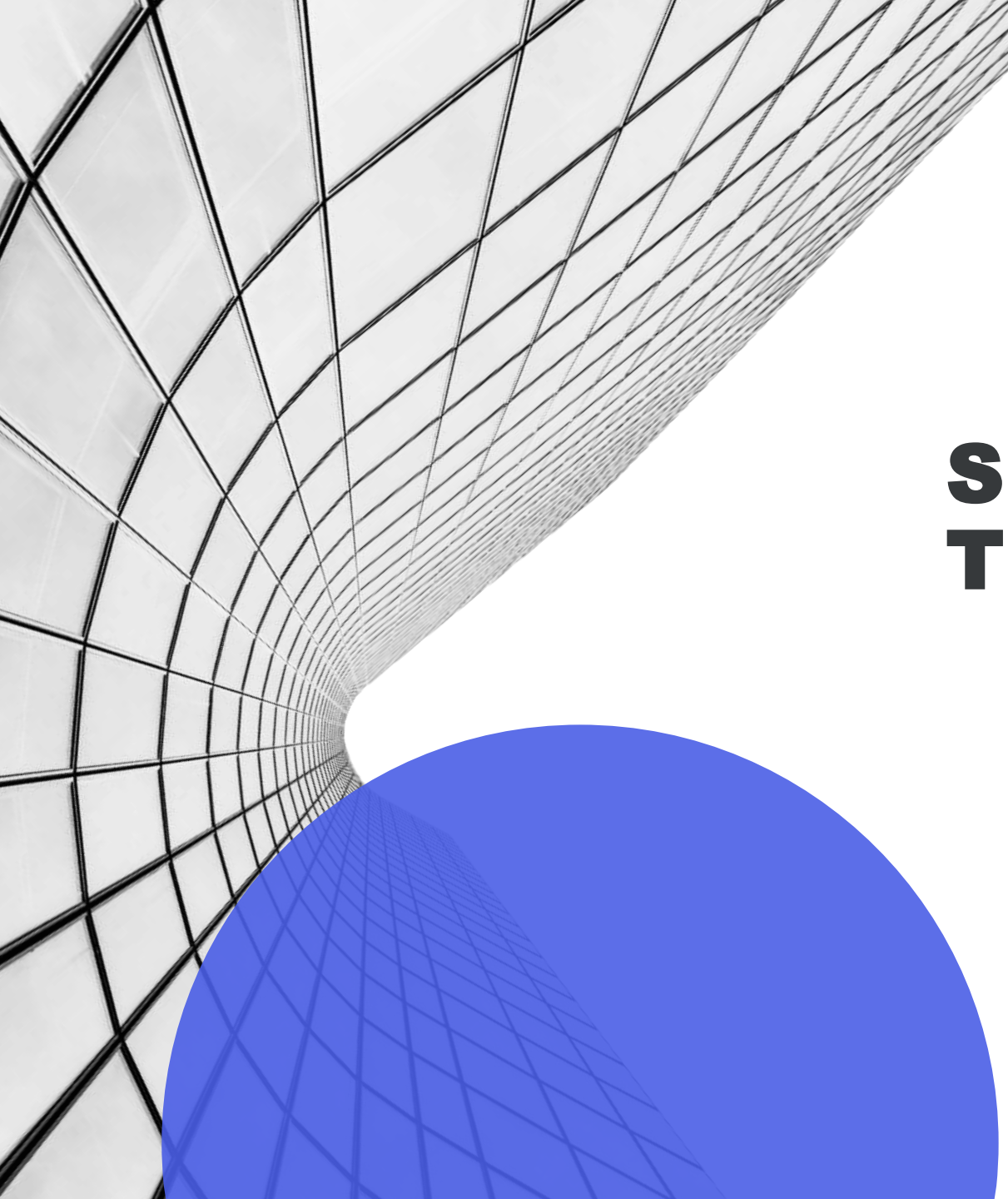
ABSTRACT



The **Real Estate Market Analysis** project is designed to understand and analyze the real estate market using property, customer, agent, and sales data. It helps find useful information such as which locations have the highest sales, which property types are most popular, which agents perform the best, and how much revenue is generated. The main goal of this project is to help businesses make better decisions, improve sales, and understand market trends more effectively.

ER-Diagram


location
Location_id INT
City VARCHAR(100)
State VARCHAR(100)
Locality VARCHAR(100)
Zip_code VARCHAR(10)
Indexes
properties
Property_id INT
Property_name VARCHAR(100)
Type_id INT
Location_id INT
Price DECIMAL(12,2)
Area_sqft INT
Bedroooms INT
Bathrooms INT
atus ENUM(...)
Listing_date DATE
Indexes
property_type
Type_id INT
Type_name VARCHAR(50)
Description VARCHAR(300)
Indexes
customers
Customer_id INT
Customer_name VARCHAR(100)
Phone VARCHAR(15)
Email VARCHAR(50)
Budget DECIMAL(12,2)
Indexes
agents
Agent_id INT
Agent_name VARCHAR(100)
Phone VARCHAR(15)
Email VARCHAR(100)
Experience INT
Commission_percent DECIMAL(5,2)
Indexes
transactions
Transaction_ID INT
Property_ID INT
Customer_ID INT
Agent_ID INT
Transaction_Date DATE
Selling_Price DECIMAL(12,2)
Payment_Mode ENUM(...)
Indexes

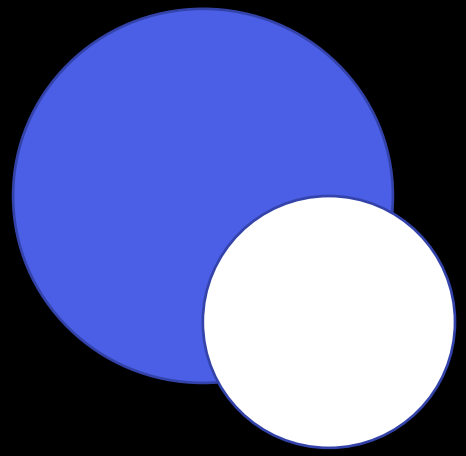


STRUCTURE OF TABLES

Property Type Table

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Type_id	int	NO	PRI	NULL	auto_increment
Type_name	varchar(50)	NO	UNI	NULL	
Description	varchar(300)	YES		NULL	



P R O P E R T Y _ T Y P E T A B L E

This table shows different categories of properties such as apartments, villas, plots, offices, and commercial spaces.

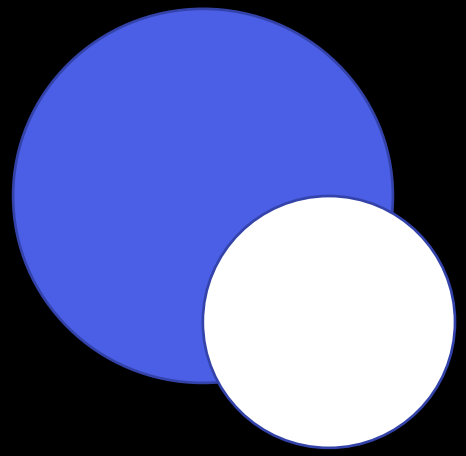
Syntax:

Desc Property_Type;

Location Table

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Location_id	int	NO	PRI	NULL	auto_increment
City	varchar(100)	NO		NULL	
State	varchar(100)	NO		NULL	
Locality	varchar(100)	NO		NULL	
Zip_code	varchar(10)	NO		NULL	



L O C A T I O N T A B L E

This table shows the location details of properties, including city, state, locality, and ZIP code.

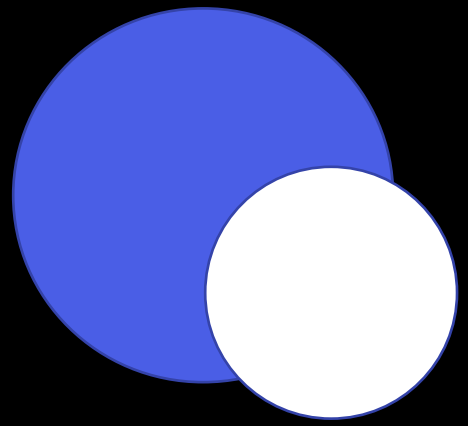
Syntax:

Desc Location;

Customers Table

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Customer_id	int	NO	PRI	NULL	auto_increment
Customer_name	varchar(100)	NO		NULL	
Phone	varchar(15)	NO	UNI	NULL	
Email	varchar(50)	NO	UNI	NULL	
Budget	decimal(12,2)	NO		NULL	



C U S T O M E R S T A B L E

This table shows the personal details and budget information of customers interested in purchasing properties.

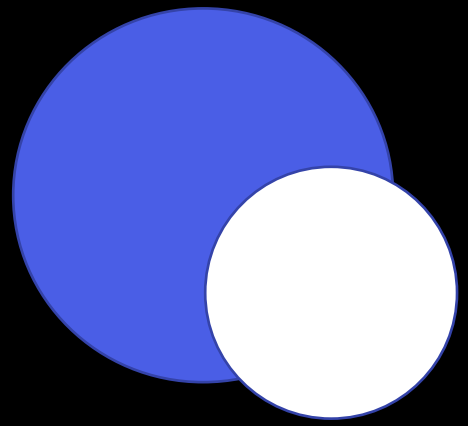
Syntax:

Desc Customers;

Agents Table

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Agent_id	int	NO	PRI	NULL	auto_increment
Agent_name	varchar(100)	NO		NULL	
Phone	varchar(15)	NO	UNI	NULL	
Email	varchar(100)	NO	UNI	NULL	
Experience	int	YES		0	
Commission_percent	decimal(5,2)	YES		2.00	



AGENTS TABLE

This table shows information about real estate agents, including their contact details, experience, and commission percentage.

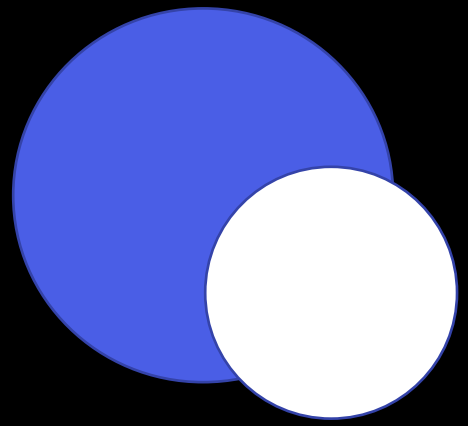
Syntax:

Desc Agents;

Properties Table

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Property_id	int	NO	PRI	NULL	auto_increment
Property_name	varchar(100)	NO		NULL	
Type_id	int	NO	MUL	NULL	
Location_id	int	NO	MUL	NULL	
Price	decimal(12,2)	NO		NULL	
Area_sqft	int	NO		NULL	
Bedroooms	int	YES		0	
Bathrooms	int	YES		0	
Status	enum('Available','Sold','Booked')	YES		Available	
Listing_date	date	NO		NULL	



PROPERTIES TABLE

This table shows complete information about each property, including its type, location, price, size, status, and listing date.

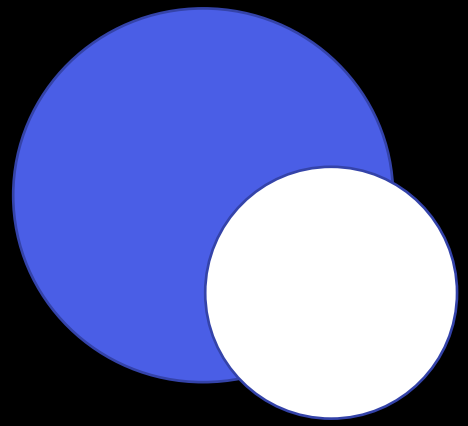
Syntax:

Desc Properties;

Transactions Table

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Transaction_ID	int	NO	PRI	NULL	auto_increment
Property_ID	int	NO	UNI	NULL	
Customer_ID	int	NO	MUL	NULL	
Agent_ID	int	NO	MUL	NULL	
Transaction_Date	date	NO		NULL	
Selling_Price	decimal(12,2)	NO		NULL	
Payment_Mode	enum('Cash','Loan','Online','Cheque')	YES		Online	

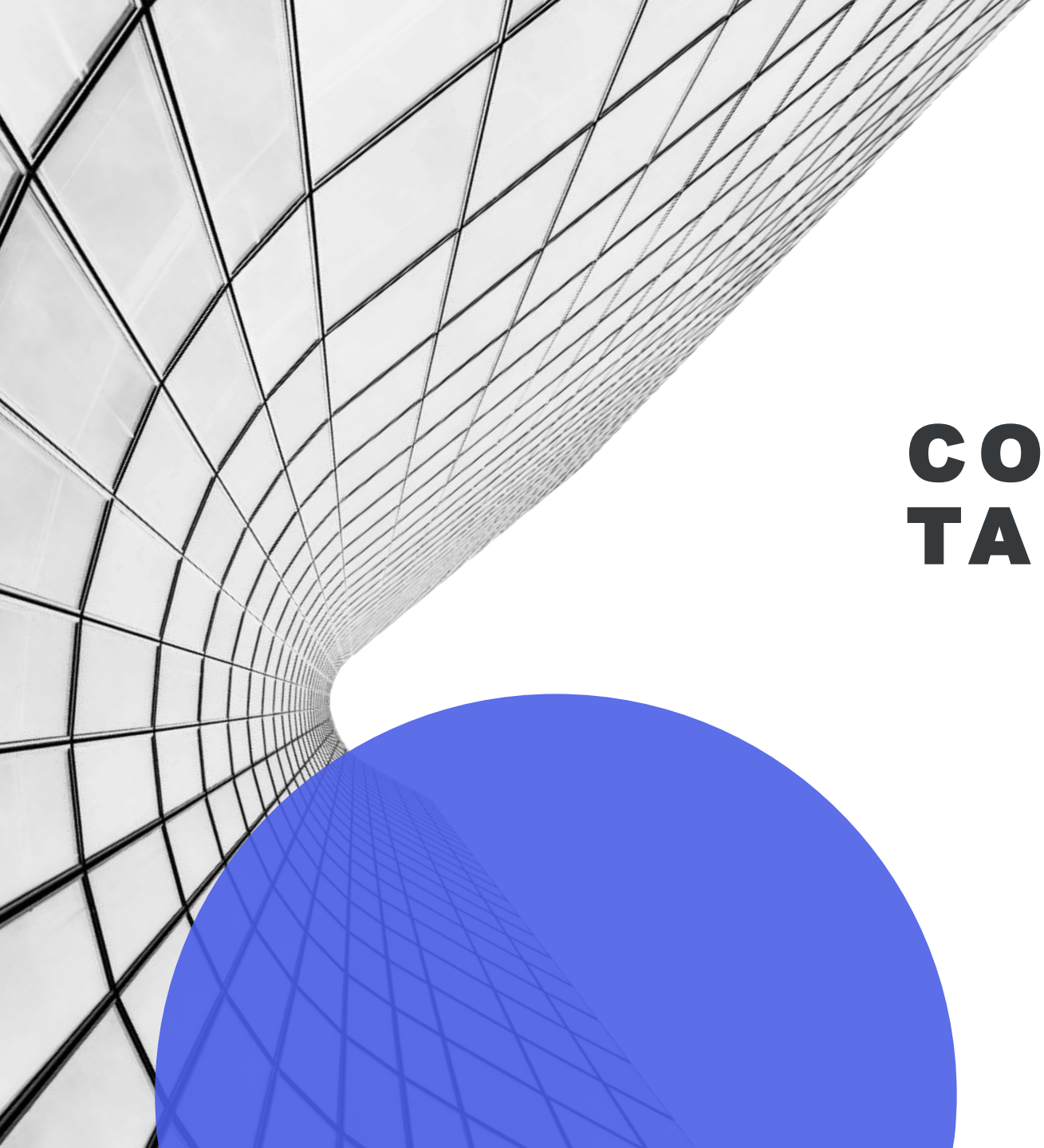


TRANSACTIONS TABLE

This table shows property sale transactions by linking customers, agents, and properties along with selling price, payment mode, and transaction date.

Syntax:

Desc Transactions;

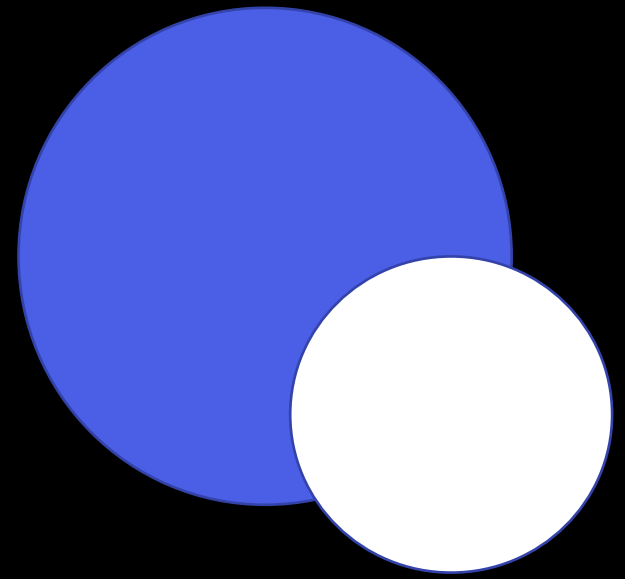


CONTENTS OF TABLE

Property Type Table

```
mysql> select * from Property_Type;
+---------+----------------------+-----------------------------------------+
| Type_id | Type_name            | Description                             |
+---------+----------------------+-----------------------------------------+
|       1 | Apartment            | Multi-storey residential apartment      |
|       2 | Villa                | Independent luxury villa                |
|       3 | Penthouse            | Luxury penthouse apartment              |
|       4 | Studio Apartment     | Compact single-room apartment           |
|       5 | Duplex               | Two-floor residential property          |
|       6 | Row House            | Connected independent residential house |
|       7 | Independent House    | Standalone residential house            |
|       8 | Residential Plot     | Land for residential construction       |
|       9 | Commercial Office    | Office space for businesses             |
|      10 | Retail Shop          | Commercial retail shop                  |
|      11 | Showroom             | Large commercial showroom               |
|      12 | Warehouse            | Storage and warehouse property          |
|      13 | Industrial Shed      | Industrial manufacturing shed           |
|      14 | Farm House           | Farmhouse with agricultural land        |
|      15 | Office Floor         | Entire office floor for commercial use  |
|      16 | Builder Floor        | Independent builder floor apartment     |
|      17 | Serviced Apartment   | Fully furnished serviced apartment      |
|      18 | Co-working Space     | Shared office workspace                 |
|      19 | Business Centre      | Commercial business center              |
|      20 | IT Park Office       | Office space in IT park                 |
|      21 | Commercial Plot      | Land for commercial development         |
|      22 | Residential Bungalow | Luxury independent bungalow             |
|      23 | Vacation Home        | Holiday residential property            |
|      24 | Mixed Use Property   | Residential and commercial combined     |
|      25 | Hostel Building      | Residential hostel property             |
|      26 | PG Accommodation     | Paying guest accommodation              |
|      27 | Shopping Complex     | Commercial shopping complex             |
|      28 | Hotel Property       | Hotel building and premises             |
|      29 | Resort Property      | Resort and leisure property             |
|      30 | Luxury Condo         | Premium condominium residence           |
+---------+----------------------+-----------------------------------------+
30 rows in set (0.00 sec)
```

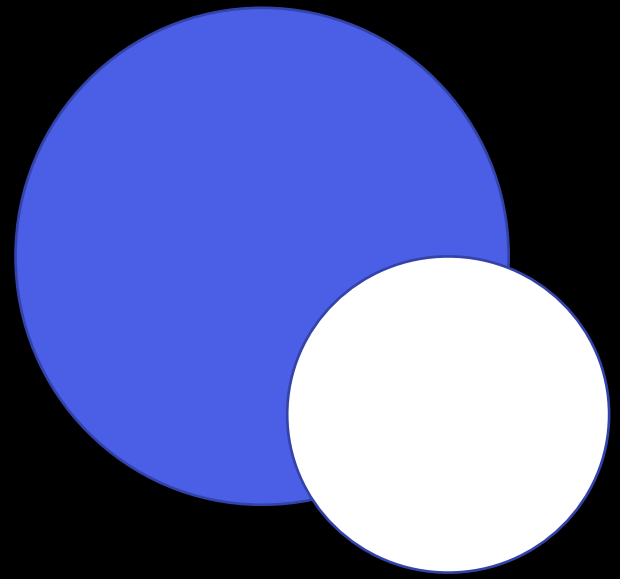
Syntax:
Select * from Property_Type;



Location Table

```
mysql> select * from Location;
+-------------+-------------+-------------+--------------------+----------+
| Location_id | City        | State       | Locality           | Zip_code |
+-------------+-------------+-------------+--------------------+----------+
|           1 | Mumbai      | Maharashtra | Bandra West        | 400050   |
|           2 | Mumbai      | Maharashtra | Andheri West       | 400053   |
|           3 | Mumbai      | Maharashtra | Powai              | 400076   |
|           4 | Mumbai      | Maharashtra | Goregaon East      | 400063   |
|           5 | Mumbai      | Maharashtra | Borivali West      | 400092   |
|           6 | Mumbai      | Maharashtra | Kandivali East     | 400101   |
|           7 | Mumbai      | Maharashtra | Chembur            | 400071   |
|           8 | Mumbai      | Maharashtra | Lower Parel        | 400013   |
|           9 | Pune        | Maharashtra | Baner              | 411045   |
|          10 | Pune        | Maharashtra | Wakad              | 411057   |
|          11 | Pune        | Maharashtra | Hinjewadi          | 411057   |
|          12 | Pune        | Maharashtra | Viman Nagar        | 411014   |
|          13 | Thane       | Maharashtra | Ghodbunder Road    | 400607   |
|          14 | Thane       | Maharashtra | Majiwada           | 400601   |
|          15 | Thane       | Maharashtra | Kasarvadavali      | 400615   |
|          16 | Thane       | Maharashtra | Manpada            | 400610   |
|          17 | Navi Mumbai | Maharashtra | Vashi              | 400703   |
|          18 | Navi Mumbai | Maharashtra | Nerul              | 400706   |
|          19 | Navi Mumbai | Maharashtra | CBD Belapur        | 400614   |
|          20 | Navi Mumbai | Maharashtra | Kharghar           | 410210   |
|          21 | Bangalore   | Karnataka   | Whitefield         | 560066   |
|          22 | Bangalore   | Karnataka   | Koramangala        | 560034   |
|          23 | Bangalore   | Karnataka   | Electronic City    | 560100   |
|          24 | Bangalore   | Karnataka   | HSR Layout         | 560102   |
|          25 | Hyderabad   | Telangana   | Hitech City        | 500081   |
|          26 | Hyderabad   | Telangana   | Gachibowli         | 500032   |
|          27 | Hyderabad   | Telangana   | Financial District | 500032   |
|          28 | Delhi       | Delhi       | Connaught Place    | 110001   |
|          29 | Delhi       | Delhi       | Saket              | 110017   |
|          30 | Delhi       | Delhi       | Dwarka             | 110075   |
+-------------+-------------+-------------+--------------------+----------+
30 rows in set (0.03 sec)
```

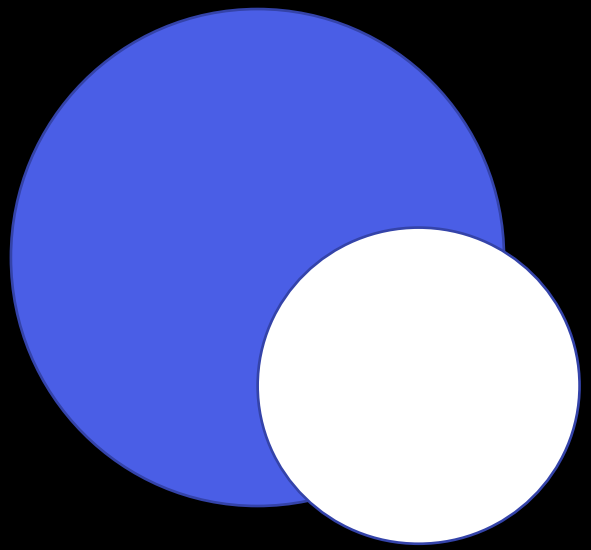
Syntax:
Select * from Location;



Customers Table

```
mysql> select * from Customers;
+-------------+------------------+------------+-----------------------------+-------------+
| Customer_id | Customer_name    | Phone      | Email                       | Budget      |
+-------------+------------------+------------+-----------------------------+-------------+
|           1 | Rajesh Gupta     | 9811000001 | rajesh.gupta@gmail.com      |  8500000.00 |
|           2 | Ananya Sharma    | 9811000002 | ananya.sharma@gmail.com     | 12500000.00 |
|           3 | Sonali Kashyap   | 9811000003 | vivek.patel@gmail.com       |  9500000.00 |
|           4 | Pooja Nair       | 9811000004 | pooja.nair@gmail.com        |  7200000.00 |
|           5 | Saniya Patil     | 9811000005 | kunal.mehta@gmail.com       | 18000000.00 |
|           6 | Sneha Joshi      | 9811000006 | sneha.joshi@gmail.com       |  6500000.00 |
|           7 | Rohan Gupta      | 9811000007 | rohan.kapoor@gmail.com      | 14000000.00 |
|           8 | Neha Verma       | 9811000008 | neha.verma@gmail.com        | 11000000.00 |
|           9 | Amit Desai       | 9811000009 | amit.desai@gmail.com        |  9000000.00 |
|          10 | Priyanka Singh   | 9811000010 | priyanka.singh@gmail.com    | 15500000.00 |
|          11 | Harsh Agarwal    | 9811000011 | harsh.agarwal@gmail.com     | 24000000.00 |
|          12 | Meera Kulkarni   | 9811000012 | meera.kulkarni@gmail.com    |  7800000.00 |
|          13 | Siddharth Rao    | 9811000013 | siddharth.rao@gmail.com     | 13000000.00 |
|          14 | Komal Shah       | 9811000014 | komal.shah@gmail.com        |  9800000.00 |
|          15 | Kajal Prasad     | 9811000015 | nitin.yadav@gmail.com       |  6200000.00 |
|          16 | Deepika Reddy    | 9811000016 | deepika.reddy@gmail.com     | 20000000.00 |
|          17 | Arjun Malhotra   | 9811000017 | arjun.malhotra@gmail.com    | 14500000.00 |
|          18 | Ritika Bose      | 9811000018 | ritika.bose@gmail.com       |  8700000.00 |
|          19 | Saurabh Jain     | 9811000019 | saurabh.jain@gmail.com      | 16500000.00 |
|          20 | Ishita Chawla    | 9811000020 | ishita.chawla@gmail.com     | 10500000.00 |
|          21 | Aditya Mishra    | 9811000021 | aditya.mishra@gmail.com     |  9200000.00 |
|          22 | Soniya Kashyap   | 9811000022 | tanvi.patil@gmail.com       |  7600000.00 |
|          23 | Rahul Khanna     | 9811000023 | rahul.khanna@gmail.com      | 17500000.00 |
|          24 | Ayesha Khan      | 9811000024 | ayesha.khan@gmail.com       | 13500000.00 |
|          25 | Varun Arora      | 9811000025 | varun.arora@gmail.com       | 11500000.00 |
|          26 | Shweta Iyer      | 9811000026 | shweta.iyer@gmail.com       |  6900000.00 |
|          27 | Manoj Bansal     | 9811000027 | manoj.bansal@gmail.com      | 22000000.00 |
|          28 | Kavya Menon      | 9811000028 | kavya.menon@gmail.com       |  8400000.00 |
|          29 | Rakesh Srivastava| 9811000029 | rakesh.srivastava@gmail.com | 27000000.00 |
|          30 | Nidhi Kapoor     | 9811000030 | nidhi.kapoor@gmail.com      | 12500000.00 |
+-------------+------------------+------------+-----------------------------+-------------+
30 rows in set (0.02 sec)
```

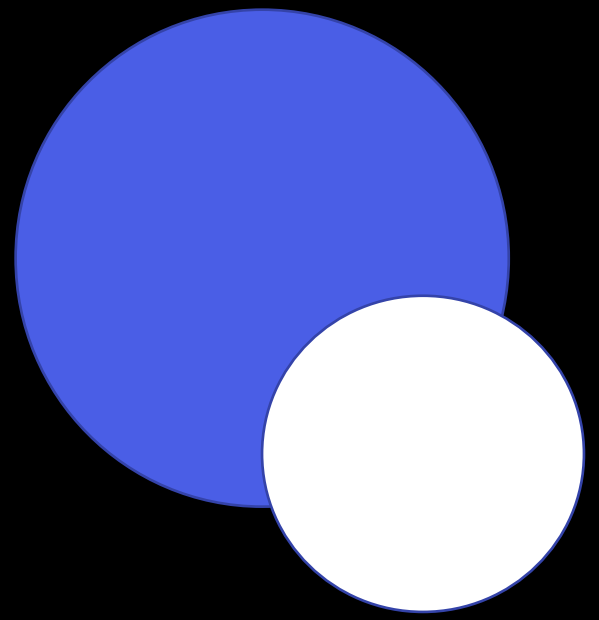
Syntax:
Select * from Customers;



Agents Table

```
mysql> select * from Agents;
+----------+----------------+------------+-------------------------------+------------+--------------------+
| Agent_id | Agent_name     | Phone      | Email                         | Experience | Commission_percent |
+----------+----------------+------------+-------------------------------+------------+--------------------+
|        1 | Rahul Sharma   | 9876501001 | rahul.sharma@urbanhomes.in    |          8 |               2.50 |
|        2 | Priya Mehta    | 9876501002 | priya.mehta@urbanhomes.in     |          6 |               2.00 |
|        3 | Amit Verma     | 9876501003 | amit.verma@urbanhomes.in      |         10 |               3.00 |
|        4 | Sneha Patil    | 9876501004 | sneha.patil@urbanhomes.in     |          5 |               2.25 |
|        5 | Rohan Desai    | 9876501005 | rohan.desai@urbanhomes.in     |          7 |               2.50 |
|        6 | Neha Kapoor    | 9876501006 | neha.kapoor@urbanhomes.in     |          9 |               2.75 |
|        7 | Vikram Singh   | 9876501007 | vikram.singh@urbanhomes.in    |         12 |               3.25 |
|        8 | Pooja Nair     | 9876501008 | pooja.nair@urbanhomes.in      |          4 |               2.00 |
|        9 | Karan Malhotra | 9876501009 | karan.malhotra@urbanhomes.in  |          6 |               2.25 |
|       10 | Anjali Joshi   | 9876501010 | anjali.joshi@urbanhomes.in    |          8 |               2.75 |
|       11 | Saurabh Gupta  | 9876501011 | saurabh.gupta@urbanhomes.in   |         11 |               3.00 |
|       12 | Meera Kulkarni | 9876501012 | meera.kulkarni@urbanhomes.in  |          5 |               2.25 |
|       13 | Arjun Reddy    | 9876501013 | arjun.reddy@urbanhomes.in     |          9 |               2.75 |
|       14 | Nikita Shah    | 9876501014 | nikita.shah@urbanhomes.in     |          3 |               1.75 |
|       15 | Deepak Yadav   | 9876501015 | deepak.yadav@urbanhomes.in    |          7 |               2.50 |
|       16 | Shweta Iyer    | 9876501016 | shweta.iyer@urbanhomes.in     |          6 |               2.25 |
|       17 | Manish Jain    | 9876501017 | manish.jain@urbanhomes.in     |         13 |               3.50 |
|       18 | Ritika Chavan  | 9876501018 | ritika.chavan@urbanhomes.in   |          4 |               2.00 |
|       19 | Abhishek Mishra| 9876501019 | abhishek.mishra@urbanhomes.in |          8 |               2.75 |
|       20 | Komal Arora    | 9876501020 | komal.arora@urbanhomes.in     |          5 |               2.25 |
|       21 | Harsh Vora     | 9876501021 | harsh.vora@urbanhomes.in      |         10 |               3.00 |
|       22 | Isha Khanna    | 9876501022 | isha.khanna@urbanhomes.in     |          7 |               2.50 |
|       23 | Nitin Bansal   | 9876501023 | nitin.bansal@urbanhomes.in    |          9 |               2.75 |
|       24 | Tanvi More     | 9876501024 | tanvi.more@urbanhomes.in      |          4 |               2.00 |
|       25 | Yash Agarwal   | 9876501025 | yash.agarwal@urbanhomes.in    |          6 |               2.25 |
|       26 | Aarti Deshmukh | 9876501026 | aarti.deshmukh@urbanhomes.in  |         11 |               3.25 |
|       27 | Rakesh Kumar   | 9876501027 | rakesh.kumar@urbanhomes.in    |         12 |               3.50 |
|       28 | Simran Kaur    | 9876501028 | simran.kaur@urbanhomes.in     |          5 |               2.25 |
|       29 | Mohit Saxena   | 9876501029 | mohit.saxena@urbanhomes.in    |          8 |               2.75 |
|       30 | Divya Rao      | 9876501030 | divya.rao@urbanhomes.in       |          7 |               2.50 |
+----------+----------------+------------+-------------------------------+------------+--------------------+
30 rows in set (0.03 sec)
```

Syntax:
Select * from Agents;



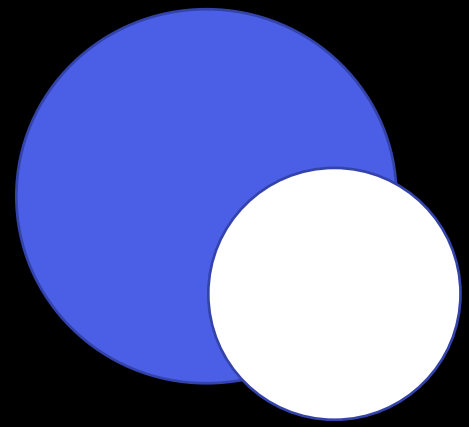
Properties Table

```
mysql> select * from properties;
```

Property_id	Property_name	Type_id	Location_id	Price	Area_sqft	Bedroooms	Bathrooms	Status	Listing_date
1	Sea View Heights A-101	1	1	18500000.00	950	2	2	Sold	2024-01-10
2	Bandra Luxury Villa	2	2	42000000.00	3200	4	4	Sold	2024-01-15
3	Powai Skyline Penthouse	3	3	68000000.00	2800	4	4	Sold	2024-02-01
4	Goregaon Studio	4	4	7200000.00	450	1	1	Sold	2024-02-05
5	Borivali Duplex Residency	5	5	24000000.00	1850	3	3	Sold	2024-02-18
6	Kandivali Row House	6	6	19500000.00	1750	3	3	Available	2024-03-01
7	Chembur Independent House	7	7	31000000.00	2400	4	3	Sold	2024-03-08
8	Lower Parel Elite Plot	8	8	55000000.00	4000	0	0	Sold	2024-03-15
9	Baner Business Office	9	9	14500000.00	1600	0	2	Sold	2024-04-01
10	Wakad Retail Hub	10	10	9800000.00	900	0	1	Sold	2024-04-06
11	Hinjewadi Grand Showroom	11	11	28000000.00	3500	0	3	Available	2024-04-12
12	Viman Nagar Warehouse	12	12	23500000.00	4200	0	2	Sold	2024-04-18
13	Ghodbunder Industrial Shed	13	13	37000000.00	5200	0	2	Sold	2024-05-01
14	Majiwada Farm House	14	14	46000000.00	6000	5	4	Booked	2024-05-08
15	Kasarvadavali Office Floor	15	15	21000000.00	2500	0	2	Sold	2024-05-16
16	Manpada Builder Floor	16	16	15800000.00	1450	3	2	Available	2024-05-25
17	Vashi Serviced Apartment	17	17	13200000.00	980	2	2	Sold	2024-06-01
18	Nerul Co-working Space	18	18	27500000.00	3000	0	4	Sold	2024-06-10
19	CBD Belapur Business Centre	19	19	52000000.00	5500	0	6	Booked	2024-06-18
20	Kharghar IT Park Office	20	20	48500000.00	5100	0	5	Sold	2024-06-28
21	Whitefield Commercial Plot	21	21	39500000.00	4500	0	0	Sold	2024-07-05
22	Koramangala Royal Bungalow	22	22	76000000.00	5000	5	5	Sold	2024-07-12
23	Electronic City Vacation Home	23	23	17500000.00	1800	3	3	Booked	2024-07-20
24	HSR Mixed Use Tower	24	24	33500000.00	2700	3	3	Available	2024-07-30
25	Hitech City Hostel Building	25	25	29500000.00	4800	20	20	Sold	2024-08-08
26	Gachibowli PG Residency	26	26	18500000.00	3500	25	25	Booked	2024-08-15
27	Financial District Shopping Complex	27	27	89000000.00	9000	0	10	Sold	2024-08-24
28	Connaught Luxury Hotel	28	28	125000000.00	12000	60	65	Booked	2024-09-01
29	Saket Green Resort	29	29	98000000.00	15000	40	42	Sold	2024-09-12
30	Dwarka Luxury Condo	30	30	32500000.00	2100	3	3	Available	2024-09-25

30 rows in set (0.02 sec)

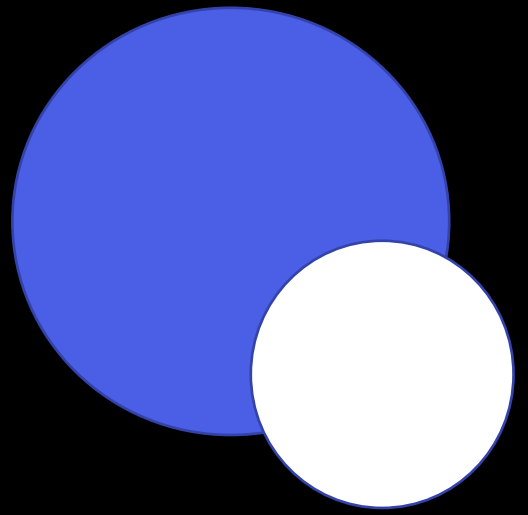
Syntax:
Select * from Properties;

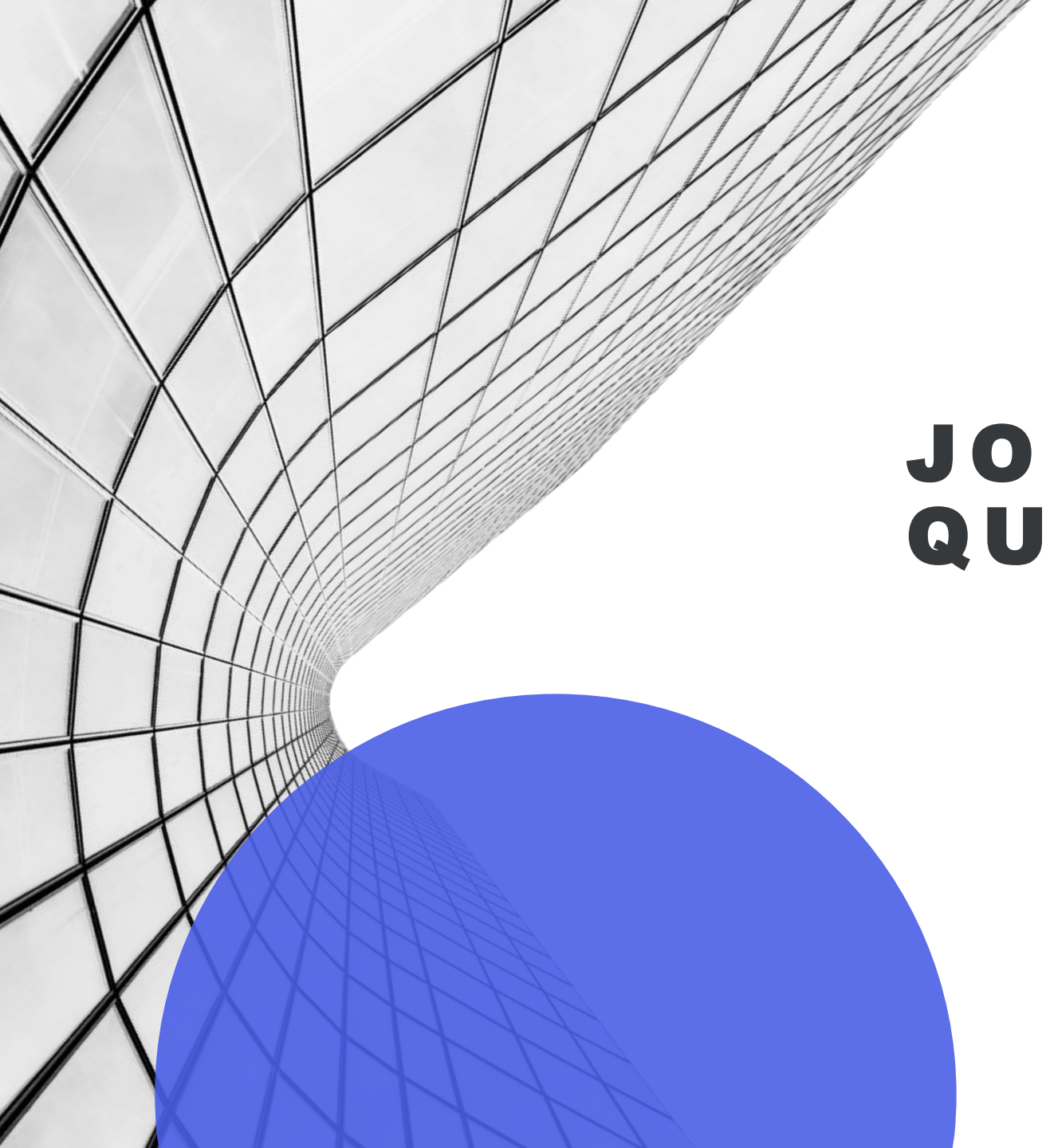


Transactions Table

```
mysql> select * from transactions;
+----------------+-------------+-------------+----------+------------------+---------------+--------------+
| Transaction_ID | Property_ID | Customer_ID | Agent_ID | Transaction_Date | Selling_Price | Payment_Mode |
+----------------+-------------+-------------+----------+------------------+---------------+--------------+
|              1 |           1 |           1 |        3 | 2024-02-15       |   18250000.00 | Loan         |
|              2 |           2 |           5 |       11 | 2024-02-28       |   41800000.00 | Online       |
|              3 |           3 |          11 |       27 | 2024-03-10       |   67500000.00 | Loan         |
|              4 |           4 |           6 |        4 | 2024-03-18       |    7100000.00 | Cash         |
|              5 |           5 |           2 |        8 | 2024-03-30       |   23800000.00 | Loan         |
|              6 |           7 |          17 |        5 | 2024-04-12       |   30800000.00 | Online       |
|              7 |           8 |          29 |       21 | 2024-04-28       |   54800000.00 | Cheque       |
|              8 |           9 |          12 |        9 | 2024-05-05       |   14350000.00 | Loan         |
|              9 |          10 |          14 |        2 | 2024-05-18       |    9650000.00 | Online       |
|             10 |          12 |          27 |       13 | 2024-06-02       |   23200000.00 | Cheque       |
|             11 |          13 |          19 |       16 | 2024-06-15       |   36500000.00 | Loan         |
|             12 |          15 |          24 |       18 | 2024-06-28       |   20800000.00 | Online       |
|             13 |          17 |          28 |        6 | 2024-07-08       |   13000000.00 | Cash         |
|             14 |          18 |          23 |       26 | 2024-07-20       |   27200000.00 | Loan         |
|             15 |          20 |          16 |        1 | 2024-08-01       |   48000000.00 | Cheque       |
|             16 |          21 |           7 |       10 | 2024-08-16       |   39000000.00 | Online       |
|             17 |          22 |          30 |       17 | 2024-08-30       |   75500000.00 | Loan         |
|             18 |          25 |          10 |       20 | 2024-09-12       |   29200000.00 | Online       |
|             19 |          27 |          25 |       15 | 2024-09-28       |   88500000.00 | Cheque       |
|             20 |          29 |          13 |       30 | 2024-10-15       |   97500000.00 | Loan         |
+----------------+-------------+-------------+----------+------------------+---------------+--------------+
20 rows in set (0.01 sec)
```

Syntax:
Select * from Transactions;





JOIN QUERIES

Inner Join

Which customers purchased a property, and which agent handled the transaction?

Customer_Name	Property_Name	Agent_Name	Selling_Price	Transaction_Date
Rajesh Gupta	Sea View Heights A-101	Amit Verma	18250000.00	2024-02-15
Saniya Patil	Bandra Luxury Villa	Saurabh Gupta	41800000.00	2024-02-28
Harsh Agarwal	Powai Skyline Penthouse	Rakesh Kumar	67500000.00	2024-03-10
Sneha Joshi	Goregaon Studio	Sneha Patil	7100000.00	2024-03-18
Ananya Sharma	Borivali Duplex Residency	Pooja Nair	23800000.00	2024-03-30
Arjun Malhotra	Chembur Independent House	Rohan Desai	30800000.00	2024-04-12
Rakesh Srivastava	Lower Parel Elite Plot	Harsh Vora	54800000.00	2024-04-28
Meera Kulkarni	Baner Business Office	Karan Malhotra	14350000.00	2024-05-05
Komal Shah	Wakad Retail Hub	Priya Mehta	9650000.00	2024-05-18
Manoj Bansal	Viman Nagar Warehouse	Arjun Reddy	23200000.00	2024-06-02
Saurabh Jain	Ghodbunder Industrial Shed	Shweta Iyer	36500000.00	2024-06-15
Ayesha Khan	Kasarvadavali Office Floor	Ritika Chavan	20800000.00	2024-06-28
Kavya Menon	Vashi Serviced Apartment	Neha Kapoor	13000000.00	2024-07-08
Rahul Khanna	Nerul Co-working Space	Aarti Deshmukh	27200000.00	2024-07-20
Deepika Reddy	Kharghar IT Park Office	Rahul Sharma	48000000.00	2024-08-01
Rohan Gupta	Whitefield Commercial Plot	Anjali Joshi	39000000.00	2024-08-16
Nidhi Kapoor	Koramangala Royal Bungalow	Manish Jain	75500000.00	2024-08-30
Priyanka Singh	Hitech City Hostel Building	Komal Arora	29200000.00	2024-09-12
Varun Arora	Financial District Shopping C...	Deepak Yadav	88500000.00	2024-09-28
Siddharth Rao	Saket Green Resort	Divya Rao	97500000.00	2024-10-15

SYNTAX:

```
select c.Customer_Name, p.Property_Name,
 a.Agent_Name, t.Selling_Price,
 t.Transaction_Date from Transactions t
inner join Customers c
on t.Customer_id=c.Customer_id
inner join Agents a
on t.Agent_id=a.Agent_id
inner join Properties p
on t.Property_id=p.Property_id;
```

select c.Customer_Name, p.Property_Name, a.Agent_Name, t.Selling_Price, t.Transaction_Date from Transactions t inner join Customers c on t.Customer_id=c.Customer_id inner join Agents a on t.Agent_id=a.Agent_id inner join Properties p on t.Property_id=p.Property_id;

Left Join

Which properties are still unsold and do not have any transaction?

Property_Name	Status	Price
Kandivali Row House	Available	19500000.00
Hinjewadi Grand Showroom	Available	28000000.00
Majiwada Farm House	Booked	46000000.00
Manpada Builder Floor	Available	15800000.00
CBD Belapur Business Centre	Booked	52000000.00
Electronic City Vacation Home	Booked	17500000.00
HSR Mixed Use Tower	Available	33500000.00
Gachibowli PG Residency	Booked	18500000.00
Connaught Luxury Hotel	Booked	125000000.00
Dwarka Luxury Condo	Available	32500000.00

SYNTAX:

```
select
p.Property_Name, p.Status,
p.Price from Properties p
left join Transactions t
on p.Property_id=t.Property_id
where t.Transaction_id is null;
```

select
p.Property_Name, p.Status,
p.Price from Properties p
left join Transactions t
on p.Property_id=t.Property_id
where t.Transaction_id is null;

Right Join

Show every agent and the property they sold (if any).

Agent_Name	Property_Name	Status	Selling_Price
Amit Verma	Sea View Heights A-101	Sold	18250000.00
Saurabh Gupta	Bandra Luxury Villa	Sold	41800000.00
Rakesh Kumar	Powai Skyline Penthouse	Sold	67500000.00
Sneha Patil	Goregaon Studio	Sold	7100000.00
Pooja Nair	Borivali Duplex Residency	Sold	23800000.00
Rohan Desai	Chembur Independent House	Sold	30800000.00
Harsh Vora	Lower Parel Elite Plot	Sold	54800000.00
Karan Malhotra	Baner Business Office	Sold	14350000.00
Priya Mehta	Wakad Retail Hub	Sold	9650000.00
Arjun Reddy	Viman Nagar Warehouse	Sold	23200000.00
Shweta Iyer	Ghodbunder Industrial Shed	Sold	36500000.00
Ritika Chavan	Kasarvadavali Office Floor	Sold	20800000.00
Neha Kapoor	Vashi Serviced Apartment	Sold	13000000.00
Aarti Deshmukh	Nerul Co-working Space	Sold	27200000.00
Rahul Sharma	Kharghar IT Park Office	Sold	48000000.00
Anjali Joshi	Whitefield Commercial Plot	Sold	39000000.00
Manish Jain	Koramangala Royal Bungalow	Sold	75500000.00
Komal Arora	Hitech City Hostel Building	Sold	29200000.00
Deepak Yadav	Financial District Shopping C...	Sold	88500000.00
Divya Rao	Saket Green Resort	Sold	97500000.00

SYNTAX:

```
select a.Agent_Name, p.Property_Name,
p.Status,t.Selling_Price
from transactions t
right join Agents a
on t.Agent_id=a.Agent_id
right join Properties p
on t.Property_id=p.Property_id
where t.Transaction_id is not null;
```

select a.Agent_Name,
p.Property_Name, p.Status,
t.Selling_Price
from transactions t
right join Agents a
on t.Agent_id=a.Agent_id
right join Properties p
on t.Property_id=p.Property_id
where t.Transaction_id is not
null;

Multiple Join

Which city generated the highest revenue, and which customers purchased properties there?

city	Customer_Name	Property_Name	Selling_price
Delhi	Siddharth Rao	Saket Green Resort	97500000.00
Hyderabad	Varun Arora	[illegible] Shopping Complex	88500000.00
Bangalore	Nidhi Kapoor	[illegible]oyal Bungalow	75500000.00
Mumbai	Harsh Agarwal	Powai Skyline Penthouse	67500000.00
Mumbai	Rakesh Srivastava	Lower Parel Elite Plot	54800000.00
Navi Mumbai	Deepika Reddy	Kharghar IT Park Office	48000000.00
Mumbai	Saniya Patil	Bandra Luxury Villa	41800000.00
Bangalore	Rohan Gupta	Whitefield Commercial Plot	39000000.00
Thane	Saurabh Jain	Ghodbunder Industrial Shed	36500000.00
Mumbai	Arjun Malhotra	Chembur Independent House	30800000.00
Hyderabad	Priyanka Singh	Hitech City Hostel Building	29200000.00
Navi Mumbai	Rahul Khanna	Nerul Co-working Space	27200000.00
Mumbai	Ananya Sharma	Borivali Duplex Residency	23800000.00
Pune	Manoj Bansal	Viman Nagar Warehouse	23200000.00
Thane	Ayesha Khan	Kasarvadavali Office Floor	20800000.00
Mumbai	Rajesh Gupta	Sea View Heights A-101	18250000.00
Pune	Meera Kulkarni	Baner Business Office	14350000.00
Navi Mumbai	Kavya Menon	Vashi Serviced Apartment	13000000.00
Pune	Komal Shah	Wakad Retail Hub	9650000.00
Mumbai	Sneha Joshi	Goregaon Studio	7100000.00

SYNTAX:

```
select l.city, c.Customer_Name,
p.Property_Name, t.Selling_price
from Transactions t
inner join Properties p
on t.Property_id=p.Property_id
inner join Customers c
on t.Customer_id=c.Customer_id
inner join Location l
on p.Location_id =l.Location_id
order by t.Selling_price desc;
```

select l.city, c.Customer_Name,
p.Property_Name,
t.Selling_price
from Transactions t
inner join Properties p
on t.Property_id=p.Property_id
inner join Customers c on
t.Customer_id=c.Customer_id
inner join Location l on
p.Location_id =l.Location_id
order by t.Selling_price desc;

Inner + Having Join

Which property types have generated more than ₹3 crore in total sales revenue?

Type_Name	Total_Sales	Total_Revenue
Resort Property	1	97500000.00
Shopping Complex	1	88500000.00
Residential Bungalow	1	75500000.00
Penthouse	1	67500000.00
Residential Plot	1	54800000.00
IT Park Office	1	48000000.00
Villa	1	41800000.00
Commercial Plot	1	39000000.00
Industrial Shed	1	36500000.00
Independent House	1	30800000.00

SYNTAX:

```
select pt.Type_Name,
COUNT(t.Transaction_ID) as Total_Sales,
SUM(t.Selling_Price) as Total_Revenue
from Property_Type pt
inner join Properties p
on pt.Type_ID = p.Type_ID
inner join Transactions t
on p.Property_ID = t.Property_ID
group by pt.Type_ID, pt.Type_Name
having SUM(t.Selling_Price) > 30000000
order by Total_Revenue desc;
```

Select pt.Type_Name,
count(t.Transaction_ID) as Total_Sales,
SUM(t.Selling_Price) as Total_Revenue
from Property_Type pt
inner join Properties p
on pt.Type_ID = p.Type_ID
inner join Transactions t
on p.Property_ID = t.Property_ID
group by pt.Type_ID, pt.Type_Name
having SUM(t.Selling_Price) > 30000000
order by Total_Revenue desc;

SUB QUERIES



Sub Queries (Average)

Property_Name	Price
Connaught Luxury Hotel	125000000.00
Saket Green Resort	98000000.00
Financial District Shopping Complex	89000000.00
Koramangala Royal Bungalow	76000000.00
Powai Skyline Penthouse	68000000.00
Lower Parel Elite Plot	55000000.00
CBD Belapur Business Centre	52000000.00
Kharghar IT Park Office	48500000.00
Majiwada Farm House	46000000.00
Bandra Luxury Villa	42000000.00
Whitefield Commercial Plot	39500000.00

WHICH PROPERTIES ARE PRICED HIGHER THAN THE AVERAGE PROPERTY PRICE?

Syntax:

```
select Property_Name,Price
from Properties where Price>(
select avg(Price) from Properties)
order by Price desc;
```

select Property_Name,Price

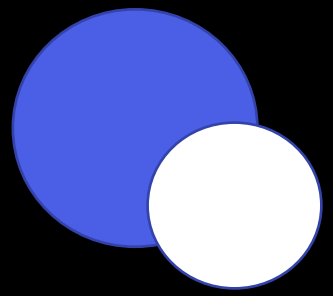
from Properties where Price>

(select avg(Price) from Properties)

order by Price desc;

Sub Queries (In)

Property_Name	Price
Sea View Heights A-101	18500000.00
Bandra Luxury Villa	42000000.00
Powai Skyline Penthouse	68000000.00
Goregaon Studio	7200000.00
Borivali Duplex Residency	24000000.00
Kandivali Row House	19500000.00
Chembur Independent House	31000000.00
Lower Parel Elite Plot	55000000.00



WHICH PROPERTIES ARE LOCATED IN MUMBAI?

Syntax:

```
select Property_Name, Price
from Properties
where Location_ID in
(select Location_ID from Location
where City = 'Mumbai');
```

select Property_Name, Price

from Properties

where Location_ID in

(select Location_ID from Location

where City = 'Mumbai');

Sub Queries (Not In)

Agent_Name	Experience
Vikram Singh	12
Meera Kulkarni	5
Nikita Shah	3
Abhishek Mishra	8
Isha Khanna	7
Nitin Bansal	9
Tanvi More	4
Yash Agarwal	6
Simran Kaur	5
Mohit Saxena	8

WHICH AGENTS HAVE NOT CLOSED A SINGLE DEAL?

Syntax:

```
select Agent_Name, Experience
from Agents
where Agent_ID not in
(select distinct Agent_ID
from Transactions);
```

select Agent_Name, Experience
from Agents
where Agent_ID not in
(select distinct Agent_ID
from Transactions);

Sub Queries (Nested)

Type_Name	Description
Apartment	Multi-storey residential apartment
Villa	Independent luxury villa
Penthouse	Luxury penthouse apartment
Studio Apartment	Compact single-room apartment
Duplex	Two-floor residential property
Independent House	Standalone residential house
Residential Plot	Land for residential construction
Commercial Office	Office space for businesses
Retail Shop	Commercial retail shop
Warehouse	Storage and warehouse property
Industrial Shed	Industrial manufacturing shed
Office Floor	Entire office floor for commercial ...
Serviced Apartment	Fully furnished serviced apartment
Co-working Space	Shared office workspace
IT Park Office	Office space in IT park
Commercial Plot	Land for commercial development
Residential Bungalow	Luxury independent bungalow
Hostel Building	Residential hostel property
Shopping Complex	Commercial shopping complex
Resort Property	Resort and leisure property

WHICH PROPERTY TYPES HAVE AT LEAST ONE SUCCESSFUL SALE?

Syntax:

```
select Type_Name,Description
from Property_Type where Type_ID in
(select distinct Type_ID
from Properties where Property_ID in
(select Property_ID from Transactions));
```

select Type_Name, Description

from Property_Type where Type_ID in

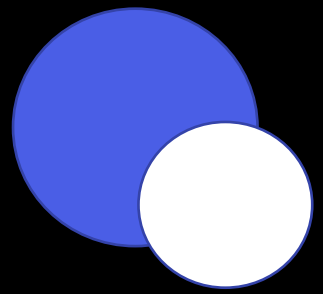
(select distinct Type_ID

from Properties where Property_ID in

(select Property_ID from Transactions));

Sub Queries (Average + Nested)

Customer_Name	Budget
Saniya Patil	18000000.00
Harsh Agarwal	24000000.00
Rakesh Srivastava	27000000.00
Deepika Reddy	20000000.00
Rohan Gupta	14000000.00
Nidhi Kapoor	12500000.00
Varun Arora	11500000.00
Siddharth Rao	13000000.00



WHICH CUSTOMERS BOUGHT PROPERTIES ABOVE THE AVERAGE SELLING PRICE?

Syntax:

```
select Customer_Name,Budget from Customers
where Customer_id in
(select Customer_id from Transactions
where Selling_Price>
(select avg(Selling_price) from Transactions ));
```

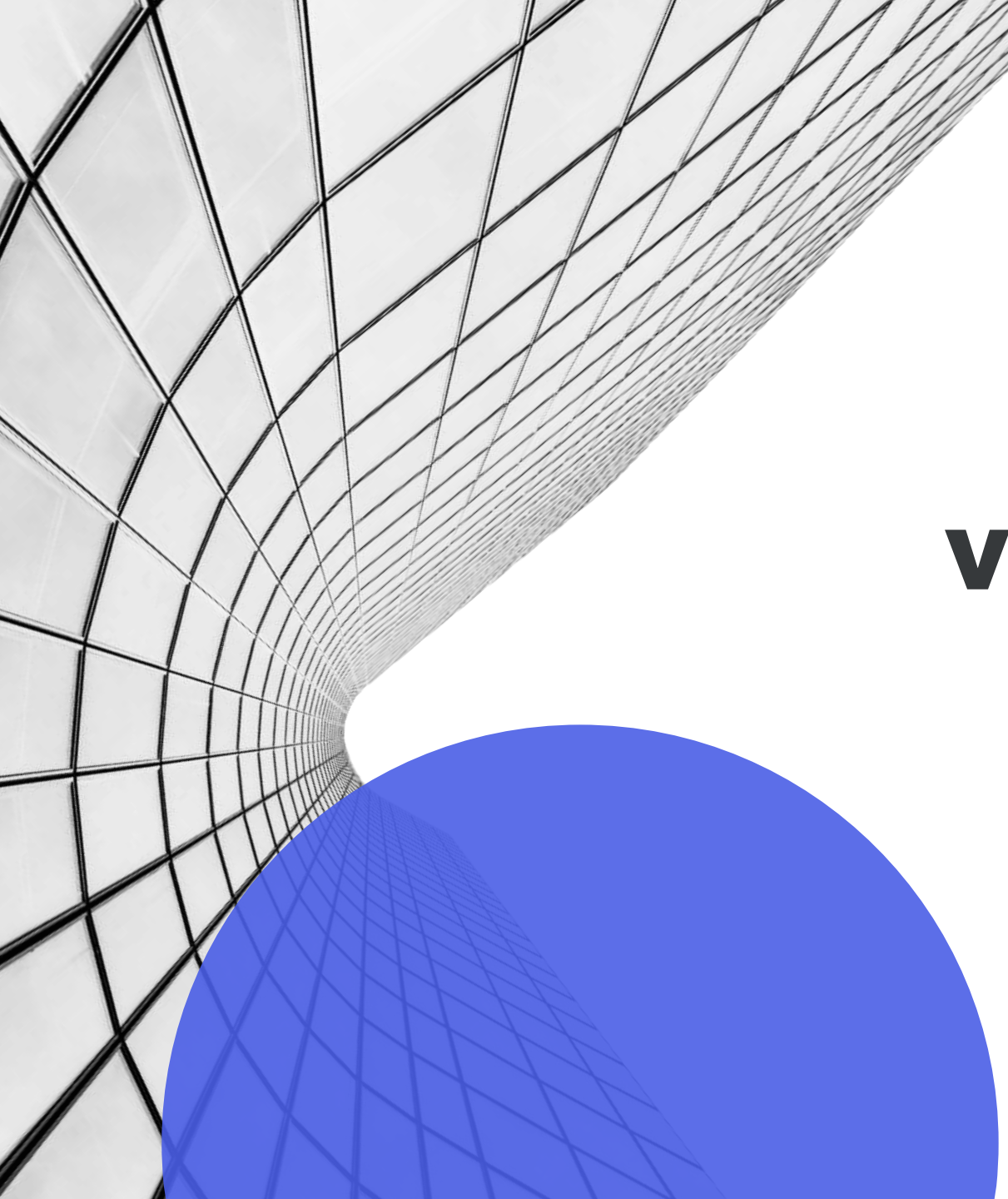
select Customer_Name,Budget from Customers

where Customer_id in

(select Customer_id from Transactions

where Selling_Price>

(select avg(Selling_price) from Transactions));



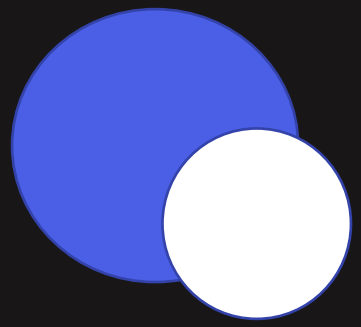
VIEWS

create one reusable view containing complete property information, including city, locality, property type, price, and status?

Property_ID	Property_Name	Property_Type	City	Locality	State	Zip_Code	Price	Area_sqft	Bedroooms	Bathrooms	Status	Listing_Date
1	Sea View Heights A-101	Apartment	Mumbai	Bandra West	Maharashtra	400050	18500000.00	950	2	2	Sold	2024-01-10
2	Bandra Luxury Villa	Villa	Mumbai	Andheri West	Maharashtra	400053	42000000.00	3200	4	4	Sold	2024-01-15
3	Powai Skyline Penthouse	Penthouse	Mumbai	Powai	Maharashtra	400076	68000000.00	2800	4	4	Sold	2024-02-01
4	Goregaon Studio	Studio Apartment	Mumbai	Goregaon East	Maharashtra	400063	7200000.00	450	1	1	Sold	2024-02-05
5	Borivali Duplex Residency	Duplex	Mumbai	Borivali West	Maharashtra	400092	24000000.00	1850	3	3	Sold	2024-02-18
6	Kandivali Row House	Row House	Mumbai	Kandivali East	Maharashtra	400101	19500000.00	1750	3	3	Available	2024-03-01
7	Chembur Independent House	Independent House	Mumbai	Chembur	Maharashtra	400071	31000000.00	2400	4	3	Sold	2024-03-08
8	Lower Parel Elite Plot	Residential Plot	Mumbai	Lower Parel	Maharashtra	400013	55000000.00	4000	0	0	Sold	2024-03-15
9	Baner Business Office	Commercial Office	Pune	Baner	Maharashtra	411045	14500000.00	1600	0	2	Sold	2024-04-01
10	Wakad Retail Hub	Retail Shop	Pune	Wakad	Maharashtra	411057	9800000.00	900	0	1	Sold	2024-04-06
11	Hinjewadi Grand Showroom	Showroom	Pune	Hinjewadi	Maharashtra	411057	28000000.00	3500	0	3	Available	2024-04-12
12	Viman Nagar Warehouse	Warehouse	Pune	Viman Nagar	Maharashtra	411014	23500000.00	4200	0	2	Sold	2024-04-18
13	Ghodbunder Industrial Shed	Industrial Shed	Thane	Ghodbunder ...	Maharashtra	400607	37000000.00	5200	0	2	Sold	2024-05-01
14	Majiwada Farm House	Farm House	Thane	Majiwada	Maharashtra	400601	46000000.00	6000	5	4	Booked	2024-05-08
15	Kasarvadavali Office Floor	Office Floor	Thane	Kasarvadavali	Maharashtra	400615	21000000.00	2500	0	2	Sold	2024-05-16
16	Manpada Builder Floor	Builder Floor	Thane	Manpada	Maharashtra	400610	15800000.00	1450	3	2	Available	2024-05-25
17	Vashi Serviced Apartment	Serviced Apartment	Navi M...	Vashi	Maharashtra	400703	13200000.00	980	2	2	Sold	2024-06-01
18	Nerul Co-working Space	Co-working Space	Navi M...	Nerul	Maharashtra	400706	27500000.00	3000	0	4	Sold	2024-06-10
19	CBD Belapur Business Centre	Business Centre	Navi M...	CBD Belapur	Maharashtra	400614	52000000.00	5500	0	6	Booked	2024-06-18
20	Kharghar IT Park Office	IT Park Office	Navi M...	Kharghar	Maharashtra	410210	48500000.00	5100	0	5	Sold	2024-06-28
21	Whitefield Commercial Plot	Commercial Plot	Bangal...	Whitefield	Karnataka	560066	39500000.00	4500	0	0	Sold	2024-07-05
22	Koramangala Royal Bungalow	Residential Bungalow	Bangal...	Koramangala	Karnataka	560034	76000000.00	5000	5	5	Sold	2024-07-12
23	Electronic City Vacation Home	Vacation Home	Bangal...	Electronic City	Karnataka	560100	17500000.00	1800	3	3	Booked	2024-07-20
24	HSR Mixed Use Tower	Mixed Use Property	Bangal...	HSR Layout	Karnataka	560102	33500000.00	2700	3	3	Available	2024-07-30

Syntax:

```
create view vw_property_listing as
select p.Property_ID, p.Property_Name,
pt.Type_Name as Property_Type,
l.City, l.Locality,
l.State, l.Zip_Code,
p.Price, p.Area_sqft,
p.Bedroooms, p.Bathrooms,
p.Status, p.Listing_Date
from Properties p
inner join Property_Type pt
on p.Type_ID = pt.Type_ID
inner join Location l
on p.Location_ID = l.Location_ID;
-- To show the view
select *
from vw_property_listing;
```



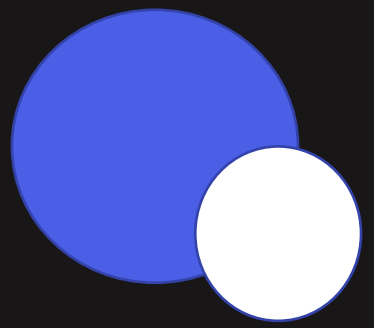
How much revenue has each agent generated and how many properties have they sold?

Agent_ID	Agent_Name	Experience	Commission_Percent	Total_Deals	Total_Revenue	Average_Sale_Value
30	Divya Rao	7	2.50	1	97500000.00	97500000.000000
15	Deepak Yadav	7	2.50	1	88500000.00	88500000.000000
17	Manish Jain	13	3.50	1	75500000.00	75500000.000000
27	Rakesh Kumar	12	3.50	1	67500000.00	67500000.000000
21	Harsh Vora	10	3.00	1	54800000.00	54800000.000000
1	Rahul Sharma	8	2.50	1	48000000.00	48000000.000000
11	Saurabh Gupta	11	3.00	1	41800000.00	41800000.000000
10	Anjali Joshi	8	2.75	1	39000000.00	39000000.000000
16	Shweta Iyer	6	2.25	1	36500000.00	36500000.000000
5	Rohan Desai	7	2.50	1	30800000.00	30800000.000000
20	Komal Arora	5	2.25	1	29200000.00	29200000.000000
26	Aarti Deshmukh	11	3.25	1	27200000.00	27200000.000000
8	Pooja Nair	4	2.00	1	23800000.00	23800000.000000
13	Arjun Reddy	9	2.75	1	23200000.00	23200000.000000
18	Ritika Chavan	4	2.00	1	20800000.00	20800000.000000
3	Amit Verma	10	3.00	1	18250000.00	18250000.000000
9	Karan Malhotra	6	2.25	1	14350000.00	14350000.000000
6	Neha Kapoor	9	2.75	1	13000000.00	13000000.000000
2	Priya Mehta	6	2.00	1	9650000.00	9650000.000000
4	Sneha Patil	5	2.25	1	7100000.00	7100000.000000

2.00

Syntax:

```sql
create view vw_agent_sales_performance as
select a.Agent_ID,
a.Agent_Name, a.Experience,
a.Commission_Percent,
count(t.Transaction_ID) as Total_Deals,
sum(t.Selling_Price) as Total_Revenue,
avg(t.Selling_Price) as Average_Sale_Value
from Agents a
inner join Transactions t
on a.Agent_ID = t.Agent_ID
group by a.Agent_ID,
a.Agent_Name, a.Experience,
a.Commission_Percent;
-- To show the view
select *
from vw_agent_sales_performance
order by Total_Revenue desc;
```



Which cities are generating the highest revenue and how many properties have been sold there?

City	Total_Properties	Total_Sales	Total_Revenue	Average_Selling_Price
Mumbai	8	7	244050000.00	34864285.714286
Hyderabad	3	2	117700000.00	58850000.000000
Bangalore	4	2	114500000.00	57250000.000000
Delhi	3	1	97500000.00	97500000.000000
Navi Mumbai	4	3	88200000.00	29400000.000000
Thane	4	2	57300000.00	28650000.000000
Pune	4	3	47200000.00	15733333.333333

Syntax:

```sql
create view vw_city_market_performance as
select l.City,
count(distinct p.Property_ID) as Total_Properties,
count(t.Transaction_ID) as Total_Sales,
coalesce(sum(t.Selling_Price), 0) as Total_Revenue,
coalesce(avg(t.Selling_Price), 0) as Average_Selling_Price
from Location l
left join Properties p
on l.Location_ID = p.Location_ID
left join Transactions t
on p.Property_ID = t.Property_ID
group by l.City;
-- To show the view
select *
from vw_city_market_performance
order by Total_Revenue desc;
```



THANK YOU!

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